

ZHAO Xuecen (Jack)

Robotics Engineer

Robot software, simulation and LiDAR safety | Python, ROS 2, Unreal Engine 5, Jetson Nano, Raspberry Pi, ESP32

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PROFESSIONAL SUMMARY

Robotics Systems Engineer (BEng Mechanical Engineering, PolyU 2026) on Stairio, an award-winning stairwell inspection robot; sole developer of its Unreal Engine 5 software twin, running the robot's own V7 Python controller against virtual CAN motor, IMU and camera devices (**579**-test suite). Systems architect of a ROS 2 robot car whose **deterministic LiDAR safety arbiter gates every motion command**, including LLM tool calls.

AWARDS

- **Gold Medal with Congratulations of the Jury**, International Exhibition of Inventions of Geneva 2026 (Stairio, project team)
- **Hong Kong first runner-up**, James Dyson Award 2026, through to the international stage (Stairio, project team)
- **Most Popular Solution Award**, PolyU Inter-departmental Final Year Project 2025/26, six departments (Vandi, team)

CORE SKILLS

Hardware: ROS 2 and SLAM (SLAM Toolbox), LiDAR, IMU and cameras (RPLiDAR A1, dual-fisheye), Raspberry Pi and Jetson (Pi Zero 2 W, Jetson Nano), Motor and servo control, ESP32 firmware (C++), Kinematics and PID (5-DOF FK/IK), SolidWorks CAD

AI: Simulation and digital twins (Unreal Engine 5), YOLO training and TensorRT (YOLO11n on Jetson Nano), OpenCV, LLM tool calling, Vision-language models, Human-in-the-loop design

Software: Python: FastAPI and Flask, Automated testing (579 simulator tests), Live video streaming, Firebase and Cloud Run

Communication: Technical writing and diagrams, Team leadership, English, Cantonese and Putonghua (IELTS 7.0)

EXPERIENCE

Robotics Systems Engineer

Mar 2026 - Present

Stairio Limited, Hong Kong

- **Software twin:** Sole developer of an Unreal Engine 5 twin in which the robot's own V7 controller (not a reimplemention) drives virtual CAN motor, IMU, environment and camera devices with fault injection in a 20-floor stairwell (319.358 m analytic rail path); **it passes all 8 acceptance-runner stages** over real HTTP, bounded by a five-level evidence plan.
- **Stream latency:** Measured the robot's live camera stream (Pi Zero 2 W, dual-fisheye, 5 fps target) in 5-minute runs: 421 ms median / 538 ms P95 local, 591 ms / 774 ms via a Cloud Run relay. Fixed a silent ~10.5 s cloud stall (the open connection hid it from socket errors) with publisher 2 s frame-ack and browser 5 s freshness reconnects; **a 90 s rerun had zero reconnects**.
- **Firmware audit:** Audited the live robot read-only (V7 edge 0.37.0, MPU6500, BME688, CANable2/C610), sending no motor command; pinned its firmware baseline by SHA-256; found the IMU sampled at 50 Hz but published at 10 Hz; **reproduced 10 firmware defects** as offline regression fixtures (e.g. a re-armed reversal dwell holding current at zero through a 5 s test).
- **IMU navigation:** Built an ideal scene-derived IMU model; replay showed the legacy floor counter missing a landing at 120 RPM as its 0.5 s gravity filter lags pitch changes. Wrote a versioned extension of the team's newer 0.91.1 detector (gyro-predicted gravity, split-corner fix, ≥ 2 s calibration before every Start) that **detects all 37 landings at 30 and 120 RPM** in replay.
- **Motion arbitration:** Built single-owner control for the Unreal console, web operator and cloud clients (session-checked commands, takeover never replays Start, 1,500 ms motion lease under a 3 s owner lease). Fixed background-tab drive stalls by moving 200 ms lease renewal into a server-side NDJSON stream, proved by **a 279 s, 105 m continuous run at 120 RPM**.
- **Video pipeline:** Streamed live 360° video from simulated cameras at the CAD lens positions in the Pi's 1280×480 dual-fisheye format, so the unmodified production operator can dewarp it (CPU shader replica: mean error 0.430/255). Hemisphere-only sampling and cached lookup maps **cut first-call panorama projection from 1630.54 to 885.65 ms** (later calls 533–570 ms, pixel-identical), so full-resolution captures finish while both live views stream.
- **Behaviour spec:** Specified Stairio 360 behaviour (not yet built): setup/routine state machine, position quality states (anchored/tracking/degraded/lost) with uncertainty, floor identity not from IMU alone, one supervisor gating all actuation.
- **Firmware and CAD:** Contributed to the Stairio S1's ESP32-P4 camera firmware (C/C++) and supported commissioning, capture review and field demonstrations; adapted parts of the team's SolidWorks robot CAD (enclosure, mounts, packaging).
- **Also:** Built the bilingual Stairio website and the Stairio HR hiring system, and made the launch video and Blender renders.

IJfreelance, Remote

- Ran an independent engineering practice for **295+** clients over four years (scoping to follow-up): SolidWorks drawings and parametric models, ANSYS design checks and 3D-printed prototypes, plus web apps and LLM workflow automation.

SELECTED PROJECTS

Vandi - Fuel-Cell-Powered Robot Car (Final Year Project)

2025 - 2026

- **Architecture:** Systems architect in a three-person, three-department team, owning all robot software and mechanical packaging; one Jetson Nano script starts **8** processes (LiDAR, TF, SLAM Toolbox, chassis, odometry, Flask, model proxy, UI).
- **Safety arbiter:** Made one deterministic LiDAR function the only path to the motors for teleop, chat, LLM tools and auto drive, since detector, autonomy and LLM can all be wrong. It reads eight 45° sectors, stops at 0.40 m and inside 0.80 m scales linear speed to a 12% floor but angular only to 35% so it can turn away; it **refuses forward motion on scans older than 0.65 s**.
- **Motion control:** Built the ROS 2 bridge (/cmd_vel if subscribed, else serial to the Yahboom board, both behind the safety function) and single-owner motion control; **the LLM never writes motor values** (ten fixed tools, max two calls per turn), and a ten-pattern guardrail rejects unsafe spoken commands.
- **Sensor fusion:** Fused vision-model boxes with 2D LiDAR scans to answer 'how far away is the chair?' **with no depth camera:** bearing = $(0.5 - \text{box centre}) \times 70^\circ$ HFOV; range = nearest return in a box width $\times 70^\circ \times 0.85$ window (6–18°), scan ≤ 1.5 s old.
- **Perception:** Trained a three-class YOLO11n sign detector on the Jetson Nano (640 px, 80 epochs, on-device TensorRT export; **0.989** mAP50 on a same-collection validation split, not field accuracy); built the HSV traffic-light colour stage and a 60 s LiDAR-scored free-drive mode that re-scans eight directions every 0.25 s.

Car Spoiler Operations - Active Spoiler Prototype

2026

- Turned an adjustable spoiler prototype into a five-stage mechatronic system (**the only full marks in the ME42011 class**): a six-link linkage on a geared positional servo, driven from the grounded-side link, under staged positional control.
- Wrote ESP32-C3 firmware (Arduino C++, PlatformIO) in which HTTP handlers only set a target and a 20 ms loop slews the servo 1° per tick inside a hard **20°–160°** clamp. One press moves one stage, Stop holds position, and the servo centres at boot.

Other Projects

- **Robot Basketball (2025):** Led an 8-member competition team; modelled the robot in SolidWorks and fixed stability issues.
- **Auto Shooter (2023):** Raspberry Pi robot; OpenCV HSV targeting drives PCA9685 pan/tilt servos; PID/PI tracking tuning.
- **5-DOF Robot Arm (2022):** CAD design; FK/IK and workspace study in simulation, then tested on the physical arm.

ADDITIONAL EXPERIENCE

Full Stack Developer

Jan 2026 - Feb 2026

Feelings AI, Hong Kong

- Built physics, camera modes and socket-server multiplayer for an AI 3D scene builder; integrated Gemini/Ollama LLM agents.
- **Also:** Designed the SQLite/Knex scene database (initialisation, migration, seeding scripts) and an undo/redo history.

AI Features Developer Intern

Dec 2025 - Jan 2026

Inspire Education Limited, Hong Kong

- Built AI personalised-learning features for the FunMA maths platform, plus Swift iOS app features.

On-site Student Assistant, CERN, Geneva, Switzerland: science tour guide explaining LHC operations

May 2019 - Jun 2019

Digital Engineering Intern, VineUp (acquired by Graduway), Gloucester, UK: helped migrate data

Apr 2019 - May 2019

Technical Intern, North Oxford BMW, Oxford, UK: engine diagnostics and performance testing

Oct 2018 - Nov 2018

EDUCATION

BEng (Hons) Mechanical Engineering

Sep 2022 - May 2026

The Hong Kong Polytechnic University

- Final year project grade A. Electives: Perceptual Robotics, Fundamentals of Robotics, Artificial Intelligence in Products.

British A-Levels and GCSE

2018 - 2022

Oxford International College, Oxford, UK. A-Levels: Mathematics, Further Mathematics, Physics, Computer Science.

References: Prof. Li-Ta HSU, Prof. AN Liang and Prof. ZHANG Dan, The Hong Kong Polytechnic University. Contact details on request.